

CTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.3163 ± 0.00275	1.109 ± 0.065	0.6248 ± 0.0297	0.3804 ± 0.00283
Cramér $\ell_2^2 \downarrow$	0.03786 ± 0.000935	0.1332 ± 0.0173	0.089 ± 0.00804	0.04274 ± 0.00114
CF-$\ell_2^2/\omega^2 \downarrow$	0.03139 ± 0.00112	0.1046 ± 0.0132	0.0759 ± 0.00752	0.03535 ± 0.00158

QTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2431 ± 0.0168	0.913 ± 0.0331	0.5733 ± 0.0173	0.2989 ± 0.0197
Cramér $\ell_2^2 \downarrow$	0.01803 ± 0.00264	0.1025 ± 0.00859	0.0805 ± 0.00429	0.0225 ± 0.00318
CF-$\ell_2^2/\omega^2 \downarrow$	0.01727 ± 0.00249	0.08651 ± 0.00763	0.07654 ± 0.00418	0.02144 ± 0.00309

ϕ TD Dirac

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1543 ± 0.0152	0.6058 ± 0.105	0.2973 ± 0.0843	0.1774 ± 0.0187
Cramér $\ell_2^2 \downarrow$	0.006702 ± 0.00142	0.0412 ± 0.015	0.01708 ± 0.00739	0.005571 ± 0.0016
CF-$\ell_2^2/\omega^2 \downarrow$	0.00582 ± 0.00131	0.03342 ± 0.0131	0.01487 ± 0.00642	0.004744 ± 0.00166

ϕ TD Cat

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2948 ± 0.0068	0.7978 ± 0.0572	0.4703 ± 0.0457	0.3277 ± 0.00858
Cramér $\ell_2^2 \downarrow$	0.02992 ± 0.00142	0.08336 ± 0.0146	0.05154 ± 0.00532	0.02996 ± 0.000945
CF-$\ell_2^2/\omega^2 \downarrow$	0.02399 ± 0.00125	0.06328 ± 0.00908	0.04426 ± 0.0053	0.02423 ± 0.000927

ϕ TD Quant

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1606 ± 0.0167	0.3775 ± 0.0669	0.1984 ± 0.0217	0.1747 ± 0.0186
Cramér $\ell_2^2 \downarrow$	0.006944 ± 0.00135	0.01208 ± 0.00505	0.009711 ± 0.00406	0.00511 ± 0.00114
CF-$\ell_2^2/\omega^2 \downarrow$	0.006277 ± 0.00126	0.01135 ± 0.00479	0.009233 ± 0.00407	0.004432 ± 0.000953